Real Numbers

- Natural Numbers

↳ Counting Numbers

↳ $N = \{1, 2, 3, 4, ---\}$

- Integers

↳ All natural numbers with their negatives including zero.

↳ $Z = \{----, -3, -2, 1, 0, 1, 2, 3, ---\}$

- Rational Numbers

↳ A number in the form of $\frac{p}{q}$ where p and q are integers and $q \neq 0$ is known as a rational number.

- Irrational Numbers

↳ Number which are not rational are known as irrational.

- Whole Number

↳ Set of Natural Numbers along with zero. is called set of whole number.

- Real Numbers system

↳ Set of both Rational and Irrational numbers is called real number system.

* $N \subset Z \subset Q \subset R$

- Inequality

Suppose a and b are two real numbers. The number a is said to be greater than b if a-b is positive and we write it symbolically as $a>b$. Again the number a is said to be less than the number b, if b-a is positive. In symbols, we write it as $a<b$.

We may sometimes write it as $b>a$ also. Here, the symbol '>' stands for 'is greater than'.

In case $b-a=0$, we say a and b are equal. we write this as $b=a$.

- Interval

↳ let a and b be two numbers on the real line. Then the set of points on the real line between a and b is known as an interval.

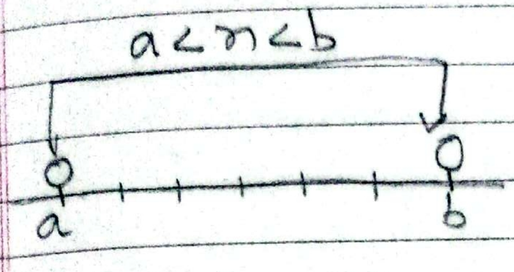
↳ Interval is denoted by I.

i) Open-interval

↳ An interval not containing the end points a and b is known as an open interval. It is denoted by (a,b).

CamScanner

i) Symbolically, $(a,b) = \{x : a < x < b\}$
The graph of the above open interval (a,b) is shown below.

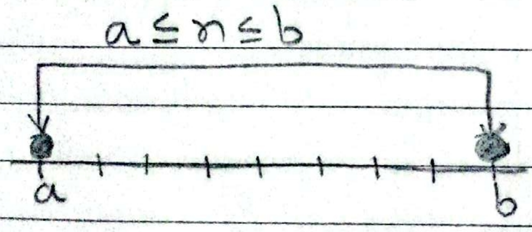


ii) Closed interval

↳ An interval containing both the end points a and b is known as a closed interval. It is denoted by $[a,b]$

Symbolically, $[a,b] = \{x : a \leq x \leq b\}$

The graph of the closed interval $[a,b]$ is shown below



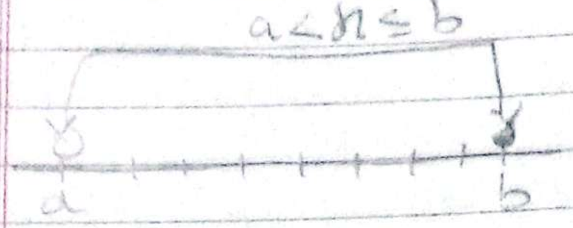
iii) Left open interval

↳ An interval not containing the end point a and containing the end point b is known as a left open interval

↳ It is denoted by $(a,b]$.

symbolically, $(a, b] = \{x : a < x \leq b\}$

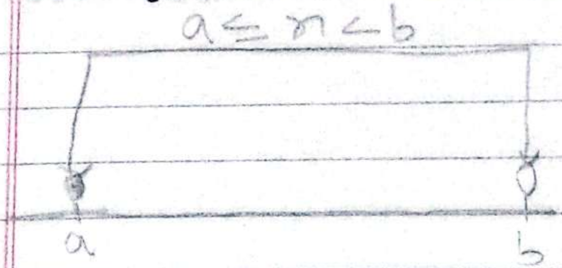
The graph of the left open interval is shown below.



iv) Right Open interval

↳ An interval containing the end point a and not containing the end point b is known as the right open interval. It is denoted by $[a, b)$.

symbolically, $[a, b) = \{x : a \leq x < b\}$
The graph of the right open interval $[a, b)$ is shown ~~aside~~ below :-



Exercise-2

1. Represent the following sets in the interval form.

a) $\{x: 4 \le x \le 7\}$
$= [4,7]$

b) $\{x: 7 < x \le 12\}$
$= (7,12]$

c) $\{x: -2 \le x < 10\}$
$= [-2,10)$

d) $\{x: x \le 5\}$
$= (-\infty,5]$

e) $\{x: \ge 10\}$
$= [10,\infty)$

f) $\{x: 0 < x < 5\}$
$= (0,5)$

2. Represent the following intervals in the set builder form

a) $[2,10]$
$= \{x: 2 \le x \le 10\}$

b) $[2,15)$
$= \{x: 2 \le x < 15\}$

c) $(-3, 12)$

$= \{x : -3 < x < 12\}$

d) $(0, 7]$

$= \{x : 0 < x \leq 7\}$

3. If $A = [0, 7]$, $B = (5, 10)$ write the following in (i) interval notation, (ii) set builder form.

b) $A \cap B$

$= [0, 7] \cap (5, 10)$

$= \{x : 0 \leq x \leq 7\} \cap \{x : 5 < x < 10\}$

$= \{x : 0 < 5 \leq 7\}$

$= (5, 7]$

a) $A \cup B$

$= [0, 7] \cup (5, 10)$

$= \{x : 0 \leq x \leq 7\} \cup \{x : 5 < x < 10\}$

$= \{x : 0 \leq x < 10\}$

$= [0, 10)$

c) $A - B$

$= [0,7] - (5,10)$

$= \{x : 0 \leq x \leq 7\} - \{x : 5 < x < 10\}$

$= \{x : 0 \leq x \leq 5\}$

$= [0,5]$

d) $A \Delta B$

$= (A-B) \cup (B-A)$

$= [0,5] \cup \{(5,10) - [0,7]\}$

$= [0,5] \cup (\{x : 5 < x < 10\} - \{x : 0 \leq x \leq 7\})$

$= [0,5] \cup (7,10)$

$= \{x : 0 \leq x \leq 5\} \cup \{x : 7 < x < 10\}$

4. For the sets $A = [-2,5)$ and $B = (0,6)$, perform the indicated operations and show them clearly in the real number line.

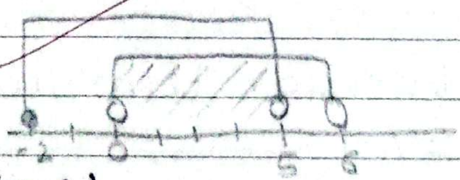
a) $A \cup B$

$= [-2,5) \cup (0,6)$

$= \{x : -2 \leq x < 5\} \cup \{x : 0 < x < 6\}$

$= [-2,6) = \{x : -2 \leq x < 6\}$

b) $A \cap B$

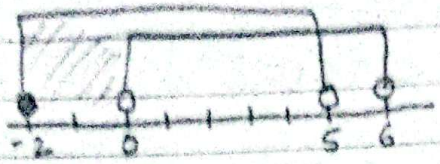


$= [-2,5) \cap (0,6)$

$= \{x : -2 \leq x < 5\} \cap \{x : 0 < x < 6\}$

$= \{x : 0 < x < 5\} = (0,5)$

c) A-B



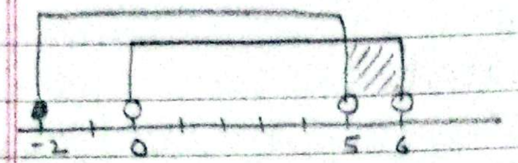
$= [-2,5) - (0,6)$

$= \{x : -2 \le x < 5\} - \{x : 0 < x < 6\}$

$= \{x : -2 \le x \le 0\}$

$= [-2,0]$

d) B-A



$= [-2,5) - (0,6)$

$= \{x : -2 \le x < 5\} - \{x : 0 < x < 6\}$

$= \{x : 5 \le x < 6\}$

$= [5,6)$

5. Solve the following inequalities.

a) $x + 5 \ge 2x - 3$

or, $5 \ge 2x - x - 3$

or, $5 + 3 \ge x$

or, $8 \ge x$

$\therefore x \le 8$ #.

b) $-3 \le x + 1 \le 7$

Subtracting 1 to each term, we get,

$-3 - 1 \le x + 1 - 1 \le 7 - 1$

$-4 \le x \le 6$

$\therefore x \in [-4, 6]$

c) $3 < 2x-1 < 9$

~~So~~ Adding 1 to each term, we get,

or, $3+1 < 2x-1+1 < 9+1$

or, $4 < 2x < 10$

Dividing each term by 2

or, $\frac{4}{2} < \frac{2x}{2} < \frac{10}{2}$

or, $2 < x < 5$

$\therefore x \in (2,5)$.

6. If $x=5$, $y=-2$, verify the following.

a) $|x+y| \leq |x|+|y|$

→ Soln,

$|x+y| = |5-2| = |3| = 3$

$|x|+|y| = |5|+|-2| = 5+2 = 7$

Hence,

$|x+y| < |x|+|y|$

verified

b) $|xy| = |x||y|$

→ Soln,

$|xy| = |5\times-2| = |-10| = 10$

$|x||y| = |5||-2| = 5\times2 = 10$

Hence,

$|xy| = |x||y| = 10$

verified.

c) $|x-y| \geq |x|-|y|$

→ Soln,

$|x-y| = |5-(-2)| = |7| = 7$

$|x|-|y| = |5|-|-2| = 5-2 = 3$

Hence,

$$|x-y| > |x| - |y|$$

verified.

d) $|x-y| \leq |x| + |y|$

$\rightarrow$ Soln,

$$|x-y| = |5-(-2)| = |5+2| = 7$$

$$|x| + |y| = |5| + |-2| = 5+2 = 7$$

Hence,

$$|x-y| = |x| + |y|$$

verified.

e) $\left|\dfrac{x}{y}\right| = \dfrac{|x|}{|y|}$

$\rightarrow$ Soln,

$$\left|\frac{x}{y}\right| = \left|\frac{5}{-2}\right| = \frac{5}{2}$$

$$\frac{|x|}{|y|} = \frac{|5|}{|-2|} = \frac{5}{2}$$

Hence,

$$\left|\frac{x}{y}\right| = \frac{|x|}{|y|} = \frac{5}{2}$$

verified.

Absolute value (Modulus)

If x be any real number then the absolute value (or modulus or numerical value) of x, written as $|x|$, is a non-negative real number defined by

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Clearly, $|x| \geq 0$. Geometrically, the absolute value of x is the distance of the point x on the real line from the origin. Moreover, the distance between any two points a and b on the real line is

$$|a-b| = |b-a|.$$

• Properties of the Absolute Value

(1) Let x be any real number.
Then
(i) $|x| \geq 0$
Proof: let $x \in R$
If $x = 0$, then $|x| = 0$
If $x > 0$, $|x| = x > 0$
If $x < 0$, $|x| = -x > 0$
$\therefore$ for all $x \in R$, $|x| \geq 0$.
proved.

ii) $|x| \geq x$ and $|x| \geq -x$

Proof: If $x \geq 0$, then $x \geq -x$

and $|x| = x \geq -x$

$\therefore |x| \geq -x$

Again,

If $x \leq 0$, then $-x \geq x$

$|x| = -x \geq x$

$\therefore |x| \geq x$

iii) $-|x| \leq x \leq |x|$

Proof: $|x| \geq x \Rightarrow x \leq |x|$

Again,

$|x| \geq -x \Rightarrow x \geq -|x|$

Combining the two results

$-|x| \leq x \leq |x|$

II. Triangle Inequality

I. Statement: For any two real numbers x and y

$$|x+y| \leq |x| + |y|$$

Proof:

$|x+y|^2 = (x+y)^2$

$= x^2 + 2xy + y^2$

$= |x|^2 + 2xy + |y|^2$

$\leq |x|^2 + 2|x||y| + |y|^2$ ($\because |x| \geq x$ and $|y| \geq y$)

$= (|x| + |y|)^2$

$\therefore |x+y| \leq |x| + |y|$

II. $|x-y| \geq |x|-|y|$

Proof:

Let $x-y=z$. Then, $x=y+z$ and

$|x| = |y+z| \leq |y| + |z|$

$|x| = |y| + |x-y|$

Hence, by transposition, we get

$|x-y| \geq |x|-|y|$.

proved

III. For any two real numbers x & y.

a) $|xy| = |x||y|$,

Proof:

For $x, y \in R$

$$|xy|^2 = (xy)^2$$
$$= x^2 y^2$$
$$= |x|^2 |y|^2$$
$$= (|x| |y|)^2$$
$$\therefore |xy| = |x||y|$$

proved.

b) $|x/y| = |x|/|y|$, $y \neq 0$

Proof:

For $x, y \in R$

$$\left|\frac{x}{y}\right|^2 = \frac{x^2}{y^2}$$
$$= \frac{|x|^2}{|y|^2}$$
$$= \left(\frac{|x|}{|y|}\right)^2$$
$$\therefore \left|\frac{x}{y}\right| = \frac{|x|}{|y|}$$

proved.

If $x \in R$ and a be any positive real number then $|x| < a \Rightarrow -a < x < a$ and conversely.

→ Proof:

For all $x \in R$, $|x| \geq x$

Given, $|x| < a$

$\therefore x \leq |x| < a \Rightarrow x < a \longrightarrow$ (i)

$\therefore$ Again for all $x \in R$, $|x| \geq -x$

~~Given $|x|<a$~~ or, $-x \leq |x|$

~~Combining (i) & (ii)~~ Given $|x| < a$

~~$-a < x < a$~~

$\Rightarrow -x < a$

$\Rightarrow x > -a$

$\Rightarrow -a < x \longrightarrow$ (ii)

Combining (i) & (ii)

$-a < x < a$

proved.

Conversely: i.e If $-a < x < a$, then $|x| < a$

Firstly, $x < a$

If $x \geq 0$ then $|x| = x$

$\therefore |x| = x < a \longrightarrow$ (i) Again,

$-a < x$

$\Rightarrow x > -a$

$\Rightarrow -x < a$

But for $x < 0$, $|x| = -x$

$\therefore |x| = -x < a \longrightarrow$ (ii)

So from (i) and (ii)

$\therefore$ For all $x \in R$, $|x| < a$ proved.

Note: For $x \in R$, $a > 0$

$|x| \leq a \Rightarrow -a \leq x \leq a$ and conversely

7. Rewrite the following using absolute value sign

a. $-2 \le x - 5 \le 6$

or, Subtracting 2 from each side

$\Rightarrow -2-2 \le x-5-2 \le 6-2$

$\Rightarrow -4 \le x-7 \le 4$

$\Rightarrow |x-7| \le 4$

b. $-7 \le x \le 0$

Multiplying each term by 2

$\Rightarrow -7\times 2 \le x \times 2 \le 0 \times 2$

$\Rightarrow -14 \le 2x \le 0$

Adding 7 on ~~both~~ side.

$\Rightarrow -14+7 \le 2x+7 \le 0+7$

$\Rightarrow -7 \le 2x+7 \le 7$

$\Rightarrow |2x+7| \le 7$

c) $6 < x+5 < 8$

Subtracting 7 on ~~but~~ each side

$\Rightarrow 6-7 < x+5-7 < 8-7$

$\Rightarrow -1 < x-2 < 1$

$\Rightarrow |x-2| < 1$

d) $4 \le 2x \le 13$

Multiplying each term by 2

$\Rightarrow 4\times 2 \le 2x \times 2 \le 13\times 2$

$\Rightarrow 8 \le 4x \le 26$

Subtracting 1[illegible] from each side,

$\Rightarrow 8-1[illegible] \le 4x-1[illegible] \le 26-1[illegible]$

$\Rightarrow -[illegible] \le 4x-1[illegible] \le$ [illegible]

$\Rightarrow -9 \le 4x-17 \le 9$

$\Rightarrow 4x-17 \leq 9$ #.

8. Rewrite the following without using absolute sign and draw its graph.

a) $|x+7|<11$

$\Rightarrow -11 < x+7 < 11$

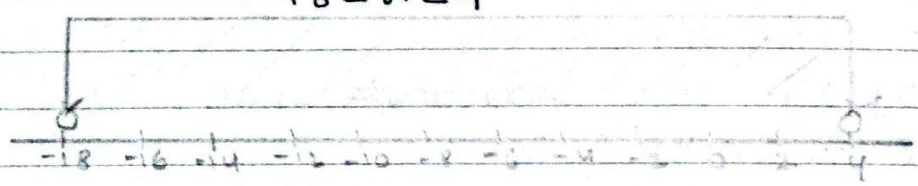
$\Rightarrow$ Subtracting 7 from each side

$\Rightarrow -11-7 < x+7-7 < 11-7$

$\Rightarrow -18 < x < 4$

$\therefore x \in (-18, 4)$

$-18<x<4$



b) $|4x+2| \leq 6$

$\Rightarrow -6 \leq 4x+2 \leq 6$

Subtracting 2 from each side

$\Rightarrow -6-2 \leq 4x+2-2 \leq 6-2$

$\Rightarrow -8 \leq 4x \leq 4$

Diving each sides by 4

$\Rightarrow \frac{-8}{4} \leq \frac{4x}{4} \leq \frac{4}{4}$

$\Rightarrow -2 \leq x \leq 1$

$\therefore x \in [-2, 1]$

$-2 \leq x \leq 1$

c) $|2x-1| \geq 3$

$\Rightarrow$ Soln,

If $(2x-1) > 0$, then $(2x-1) \geq 3$

$\Rightarrow 2x-1 \geq 3$

$\Rightarrow 2x-1+1 \geq 3+1$

$\Rightarrow 2x \geq 4$

$\Rightarrow x \geq 2$

$\Rightarrow x \in [2, \infty)$

Again, if $(2x-1) < 0$ then

$\Rightarrow -(2x-1) \geq 3$

$\Rightarrow 2x-1 \leq -3$

$\Rightarrow 2x-1+1 \leq -3+1$

$\Rightarrow 2x \leq -2$

$\Rightarrow x \leq -1$

$\Rightarrow x \in (-\infty, -1]$

$\therefore$ The required solution is $x \in (-\infty, -1] \cup [2, \infty)$

$x \in (-\infty, -1] \cup [2, \infty)$

d) $|x-1| > 1$

$\Rightarrow$ Soln,

If $(x-1)$ is > 0, then $x-1 > 1$

$\Rightarrow x-1 > 1$

$\Rightarrow x-1+1 > 1+1$

$\Rightarrow x > 2$

$\Rightarrow x \in (2, \infty)$

Again, If $(x-1) < 0$ then,

$\Rightarrow -(x-1) > 1$

$\Rightarrow x-1 < -1$

$\Rightarrow x-1+1 < -1+1$

$\Rightarrow x < 0$

$\Rightarrow x \in (-\infty, 0)$

9. If x, y and z are elements from the set of real numbers prove that:

a) $|x-y| \leq |x| + |y|$

$\rightarrow$ Proof:

$$|x-y|^2 = (x-y)^2$$
$$= x^2 - 2xy + y^2$$
$$= |x|^2 + 2(-x)y + |y|^2$$
$$\leq |x|^2 + 2|x||y| + |y|^2$$
$$(\because |x| \geq -x \text{ and } |y| \geq y)$$
$$|x-y|^2 = (|x|+|y|)^2$$
$$|x-y| = |x| + |y|$$
$$\therefore |x-y| \leq |x| + |y|$$

proved.

b) $|x-y| < a \Rightarrow y-a < x < y+a$

$\rightarrow$ Proof:

$$|x-y| < a$$

$\Rightarrow -a < x-y < a$

$\Rightarrow$ Adding y on each side

$\Rightarrow -a+y < x-y+y < a+y$

$\Rightarrow y-a < x < y+a$

proved.

c) $|x-y| + |y+z| \geq |x-z|$

$\rightarrow$ Proof:

let $(x-y) = a \rightarrow$ (i)

and $(y-z) = b \rightarrow$ (ii)

$\Rightarrow |x-y| = |a|$ and $|y-z| = |b|$

Now, adding (i) and (ii)

$x-y+y-z = a+b$

$\Rightarrow x-z = a+b$

$\Rightarrow |x-z| = |a+b|$

$\leq |a| + |b|$ [using Δ inequality]

$\leq |x-y| + |y-z|$

$\therefore |x-y| + |y-z| \geq |x-z|$

proved.

10. Solve

b). $x^2+7x+10<0$

→ Soln,

The corresponding equation of the given in equation is

$x^2+7x+10=0$

or, $x^2+5x+2x+10=0$

or, $x(x+5)+2(x+5)=0$

or, $(x+2)(x+5)=0$

$\therefore x=-2,-5$

So there are three possible intervals which are $(-\infty,-5)$, $(-5,-2)$, $(-2,\infty)$. Also the sign of $x^2+7x+10$ in these intervals are as follows:

Interval	Sign of		
	$x+2$	$x+5$	$(x+5)(x+2)$
$-\infty$ to -5	$-$	$-$	$+$
-5 to -2	$-$	$+$	$-$
-2 to ∞	$+$	$+$	$+$

From the above table, the possible interval is $(-5,-2)$.

$\therefore$ The required solution is $x\in(-5,-2)$.

a) $x^2-5x>0$

→ Soln,

Corresponding eq^n of the given eq^n is

$x^2-5x=0$

$\Rightarrow x(x-5)=0$

$\Rightarrow \therefore x=0,5$

So, there are three possible intervals which are $(-\infty, 0)$, $(0, 5)$, $(5, \infty)$. Also the sign of x^2-5x in these intervals are as follows:

Interval	Sign		
	x	$(x-5)$	$x(x-5)$
$-\infty$ to 0	$-$	$-$	$+$
0 to 5	$+$	$-$	$-$
5 to ∞	$+$	$+$	$+$

From above table, the possible interval ~~is~~ are $(-\infty, 0)$ and $(5, \infty)$

$\therefore$ Required solution is $x \in (-\infty, 0) \cup (5, \infty)$

c) $4+3x-x^2 \geq 0 \Rightarrow x^2-3x-4 \leq 0$

$\rightarrow$ Soln,

The corresponding eqn of the given equation is

$4+3x-x^2=0$

or, $x^2-3x-4=0$

or, $x^2-4x+x-4=0$

or, $x(x-4)+1(x-4)=0$

or, $(x+1)(x-4)=0$

$\therefore x=-1, 4$

So there are three possible intervals which are $(-\infty, -1]$, $[-1, 4]$, $[4, \infty)$. Also the sign of $4-3x-x^2$ are as follow

Interval	Sign		
	$(x+1)$	$(x-4)$	$(x+1)(x-4)$
$-\infty$ to -1	$-$	$-$	$+$
-1 to 4	$+$	$-$	$-$
4 to ∞	$+$	$+$	$+$

From the above table, the possible interval is $[-1, 4]$

$\therefore$ Required Solution is $[-1, 4]$ #.

d) $\dfrac{x(x-2)}{x+1} \leq 0$

$\rightarrow$ Soln,

Here the possible intervals are $(-\infty, -1)$, $(-1, 0)$, $(0, 2)$, $(2, \infty)$. Also the sign of $\dfrac{x(x-2)}{x+1}$ are as follow

Interval	Sign			
	x	$(x-2)$	$(x+1)$	$\dfrac{x(x-2)}{x+1}$
$-\infty$ to -1	$-$	$-$	$-$	$-$
-1 to 0	$-$	$-$	$+$	$+$
0 to 2	$+$	$-$	$+$	$-$
2 to ∞	$+$	$+$	$+$	$+$

From the above table the possible intervals are $(-\infty, -1) \cup (0, 2)$

$\therefore$ Required solution are $(-\infty, -1) \cup (0, 2)$ #.

- MCQ's

1. The closed interval between the numbers 1 and 5 represented by $[1,5]$ is

a) $[1,5] = \{x : 1 < x < 5\}$
b) $[1,5) = \{x : 1 \le x < 5\}$
c) $(1,5] = \{x : 1 < x \le 5\}$
d) $[1,5] = \{x : 1 \le x \le 5\}$

2. The interval $(0,7]$ can be represented in set builder form as.

a) $(0,7] = \{x : 0 \le x \le 7\}$
b) $(0,7] = \{x : 0 < x \le 7\}$
c) $(0,7] = \{x : 0 < x < 7\}$
d) $(0,7] = \{x : 0 \le x < 7\}$

3. The inequality $-1 \le x \le 7$ is in absolute value sign as

a) $|x-2| \le 4$
b) $|x-3| \le 4$
c) $|x-1| \le 2$
d) $|x-5| \le 2$

4. The solution set of the inequality $|2x-5| \le 7$ is

a) $\{x : -1 < x < 6\}$
b) $\{x : -1 \le x \le 6\}$
c) $\{x : -1 \le x < 6\}$
d) $\{x : -1 < x \le 6\}$

5. The Solution set of the inequality $|2x+1| \ge 3$ is

a) $(-\infty, -2) \cup (1, \infty)$
b) $(-\infty, -2] \cup [1, \infty)$
c) $[-\infty, -2] \cup [1, \infty]$
d) $[-\infty, -2] \cup (1, \infty]$

6. If $A = [-2,3]$ and $B = (1,5]$, then $B - A$ is

a) $[-2,1]$
b) $(1,3)$
c) $[1,3]$
d) $(3,5]$